

Name: _____

In each triangle, the drawn segment is parallel to the third side.

1. The pieces of one side are 6 and 9 (vertex first). The upper piece of the other side is 8. Find the lower piece.

Parallel or not? Use the Converse.

2. Pieces 10, 14 on one side and 15, 21 on the other.
3. Pieces 6, 9 on one side and 10, 14 on the other.

Three parallel lines cut two transversals. (Figure at right.)

4. The left pieces are 12 and 18; the upper right piece is x and the lower right is 24. Find x .
5. The left pieces are 20 and x ; the right pieces are 25 and 35. Find x .

Angle bisectors.

6. In $\triangle ABC$, $\overline{BD}$ bisects $\angle B$, $AB = 21$, $BC = 14$, and $AC = 25$. Find AD and DC .
7. In $\triangle JKL$, $\overline{KM}$ bisects $\angle K$, $JM = x$, $ML = x - 4$, $JK = 24$, and $KL = 16$. Find x and JL .

Mixed.

8. In $\triangle ABC$, $\overline{XY} \parallel \overline{BC}$, $AX = 9$, $AY = 12$, and $YC = 8$. Find XB and AB .
9. Three parallel streets cross two straight avenues. Between the streets, one avenue is cut into 240-ft and 300-ft blocks; the other's first block is 200 ft. Find its second block.
10. Complete: an angle bisector of a triangle divides the opposite side in the same ratio as the _____.

